Bangladesh University of Engineering and Technology

ME 310- Thermo-Fluid System Design

Project Title: ***Solar collector Thermosyphon Type Double Pipe Heat exchanger***

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Abstract

In this project, our objective was to use solar energy to preheat water from 8 °C to 12 °C using a solar collector Thermosyphon and a DPHX. Since, the energy exchange was not high, DPHX was chosen over more advanced heat exchangers to optimize cost. We investigated the performance of the DPHX where refrigerant R123 flows through the inner tube and water through the annular region. R123 enters the heat exchanger at 32°C and exit at 22°C, undergoing phase change as its boiling point is 26.7°C. Here water, acting as the cooling medium, enters the

annular region at 8°C. The mass flow rates of R123 and water are 0.01 kg/s and 0.1 kg/s, respectively.

Here, longitudinal fins are added on the outer surface of the inner tube to enhance heat exchanging surface and create turbulence on the fluid. To assess the effectiveness of fins, we calculated the required lengths for both finned and unfinned heat exchangers to achieve the target heat transfer. It was visible that with fin the required length was sufficiently lower than the length of unfinned condition. Moreover, due to phase change heat transfer, there is no need to use pump to circulate the hot fluid(Refrigerant-123).Hot fluid will naturally circulate because buoyancy effect. Pressure drop was calculated to determine the required pumping power. The physical model was designed at SolidWorks. Problem was modeled at HTRI and hand calculation was done to validate the HTRI result. After calculation the DPHX was fabricated which included buying materials (pipe, socket, insulation etc.) and welding them in a workshop.

Introduction

Renewable energy is essential for building a sustainable development. It helps reduce greenhouse gas emissions, combat climate change, and promote the use of clean, affordable energy (aligned with SDGs 7 and 13). Among renewable energy applications, solar energy has gained significant attention, particularly for water heating systems. Traditional methods like fuel burning for water heating contribute to air pollution, carbon emissions, and rising global temperatures. Solar energy offers a cleaner and more sustainable alternative, helping to lower energy costs and environmental impact.

In our project, we focus on designing and evaluating a **thermosyphon-type solar water heating system** using a Double Pipe Heat Exchanger (DPHX). The system is designed to preheat water from 8°C to 12°C by utilizing solar energy. A thermosyphon system efficiently transfers heat without requiring pumps or additional power, because of phase change heat transfer and buoyancy effect, making it a cost-effective and environmentally friendly solution. The DPHX is chosen for its simple design and cost effectiveness. Solar heater produce moderate heat. As a result there is a moderate energy saving. DPHX is a good option due to cost effectiveness.

A DPHX, often referred to as a hairpin heat exchanger, consists of two concentric pipes:

1. Inner pipe (called the "pipe")
2. Outer pipe (called the "annulus")

One fluid flows through the inner pipe, while the other flows through the annular region. Heat transfer will occur in two configurations:

1.Parallel flow (both fluids flow in the same direction)

2.counter flow (fluids flow in the opposite directions) For our project, the counter flow configuration is used as it provides a higher temperature difference, improving the heat transfer efficiency.

One limitation of the DPHX is its relatively small heat transfer area.

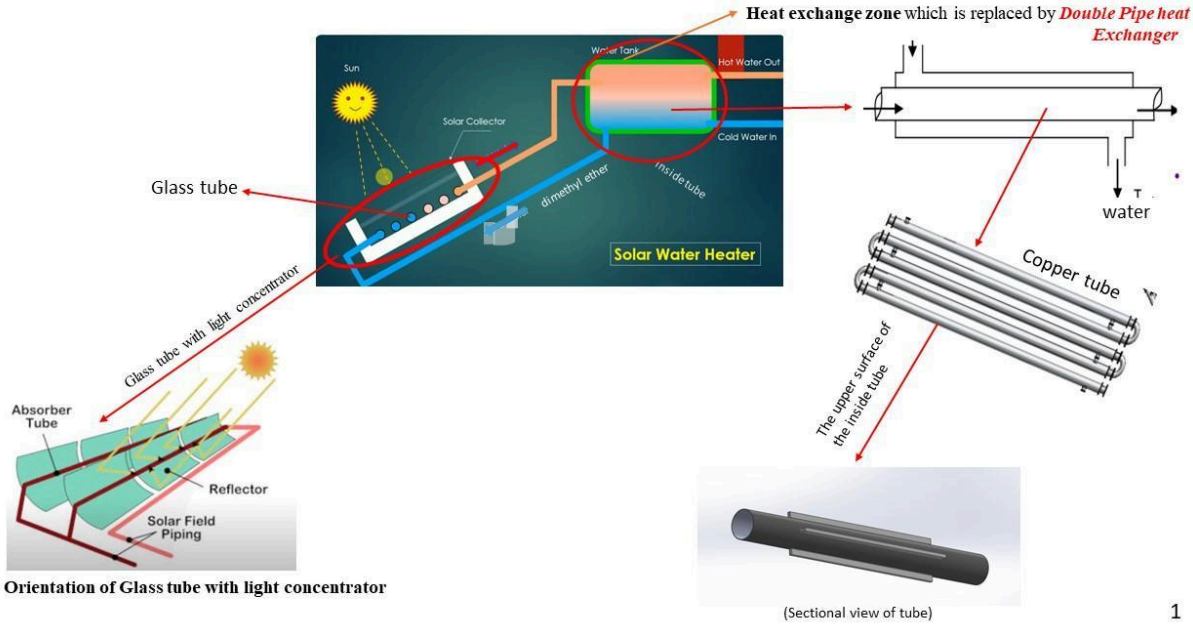
To overcome this, **longitudinal fins** are added to the outer surface of the inner pipe. These fins increase the heat transfer area without causing significant resistance to fluid flow or large pressure drops.These fins also increase turbulence for better heat transfer. As a result, the overall heat transfer rate is enhanced while maintaining system efficiency.

Our project involves designing the DPHX in SolidWorks, analyzing its performance using HTRI software, and validating the results through hand calculations. By integrating a thermosyphon solar heating system with a DPHX, this study demonstrates a cost-effective and environmentally friendly solution for preheating water, showcasing the potential of renewable energy in practical applications.

Problem Statement:

R123 enters at 32°C into the tube of the double pipe heat exchanger and exit from the tube at 22°C. The boiling point of R123 is 26.7°C.The mass flow rate of R123 into the tube will be 0.01 kg/s. Water enters into the annular region of the double pipe heat exchanger at 8°C with a mass flow rate of 0.1 kg/s.

Solar collector Thermosyphon Type Double Pipe Heat exchanger



Orientation of Glass tube with light concentrator

1

Framework

Objectives

The objectives of the project are to –

- Enhance thermal performance
- Incorporate longitudinal fin and observe the changes in result due to this
- Determine the pressure drop and keep it under critical value
- Find the optimum and economical design for the heat exchanger
- Manufacture a compact heat exchanger
- To increase the overall efficiency of a double pipe heat exchanger
- Easy installation and repairable design

Main Features

- ✔ Hot Refrigerant(R-123) will flow through the tube.
- ✔ Cold water will be on annular side.
- ✔ Longitudinal fin is connected to the outer part of the inner tube.
- ✔ Fin dimensions will be considered for deriving various parameters like equivalent diameters, hydraulic diameters ,overall heat transfer area etc.
- ✔ Overall heat transfer coefficient will be observed with different fin dimensions and numbers.
- ✔ Pressure drop variation with number and height of fins will be inspected

Operating Principles

- Hot fluid transfers heat to cold fluid through the pipe walls.
- Fins attached to pipe increases heat transfer area as well as turbulence.
- Thermal contact resistance should be eliminated between fins and tube.
- The materials should have high thermal conductivity.

- Counter flow between the two fluids will be preferred for highly constant temperature difference throughout the length of the pipe.
- Proper thermal insulation should be used at outer pipe.

Advantages of Longitudinal Fin

I. Enhanced Heat Transfer: Longitudinal fins increase the surface area for heat exchange, improving thermal performance.
II. Reduced Pipe Length: Improved efficiency reduces the required length of the heat transfer surface, saving space and material.
III. Cost-Effective Design: Decreased material usage and compact design lower manufacturing and installation costs.
IV. Versatility: Suitable for various applications involving heat exchangers, cooling, or heating systems.
V. Improved Energy Efficiency: Facilitates better energy utilization, reducing operating costs over time.

Applications:

Chemical Processing Industries: Used in reactors and distillation units for efficient heat transfer in compact spaces.

HVAC Systems: Enhances cooling or heating in air conditioning and refrigeration systems.

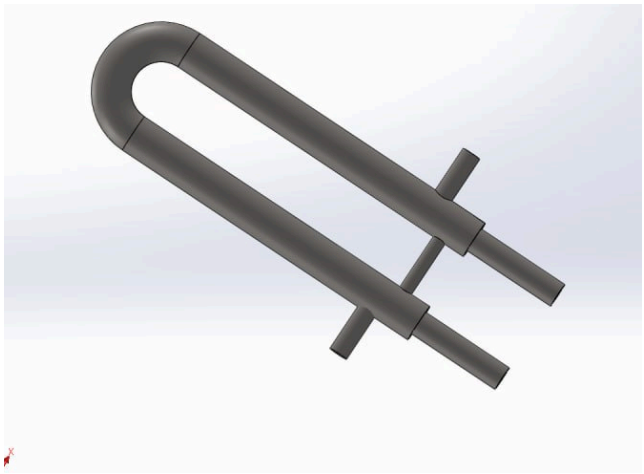
Power Plants: Facilitates heat recovery in condensers and economizers.

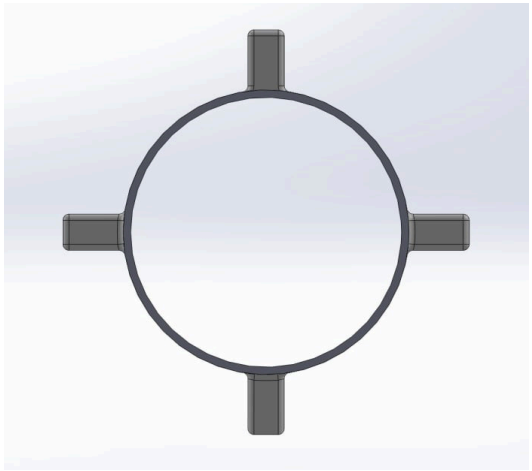
Oil and Gas Industry: Improves heat exchange in crude oil preheaters and gas processing systems.

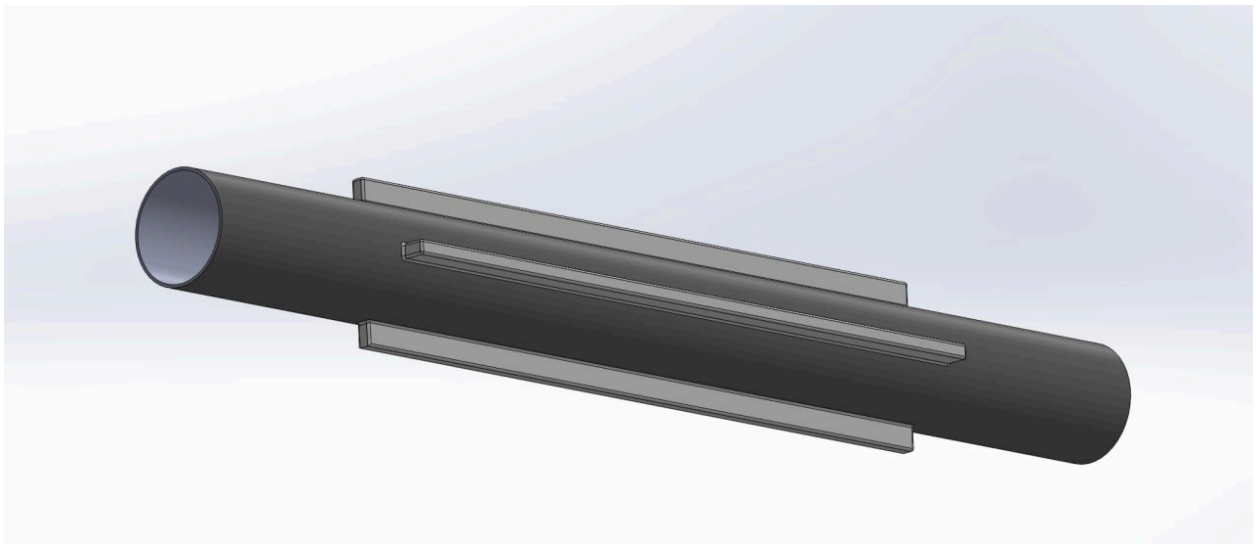
Food and Beverage Industry: Ensures precise temperature control during heating or cooling of liquids.

Renewable Energy Systems: Applied in solar water heaters and geothermal systems for improved efficiency.

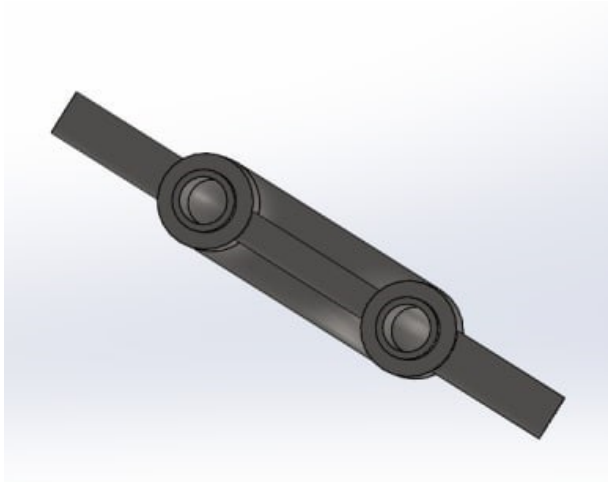
3D Design



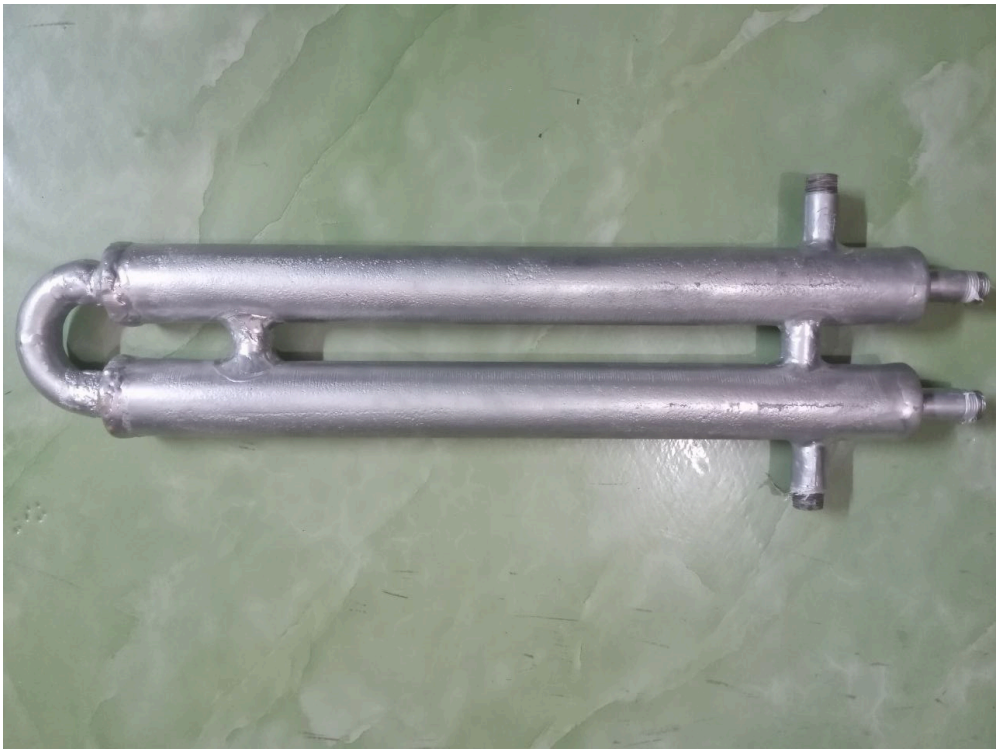








Final Design



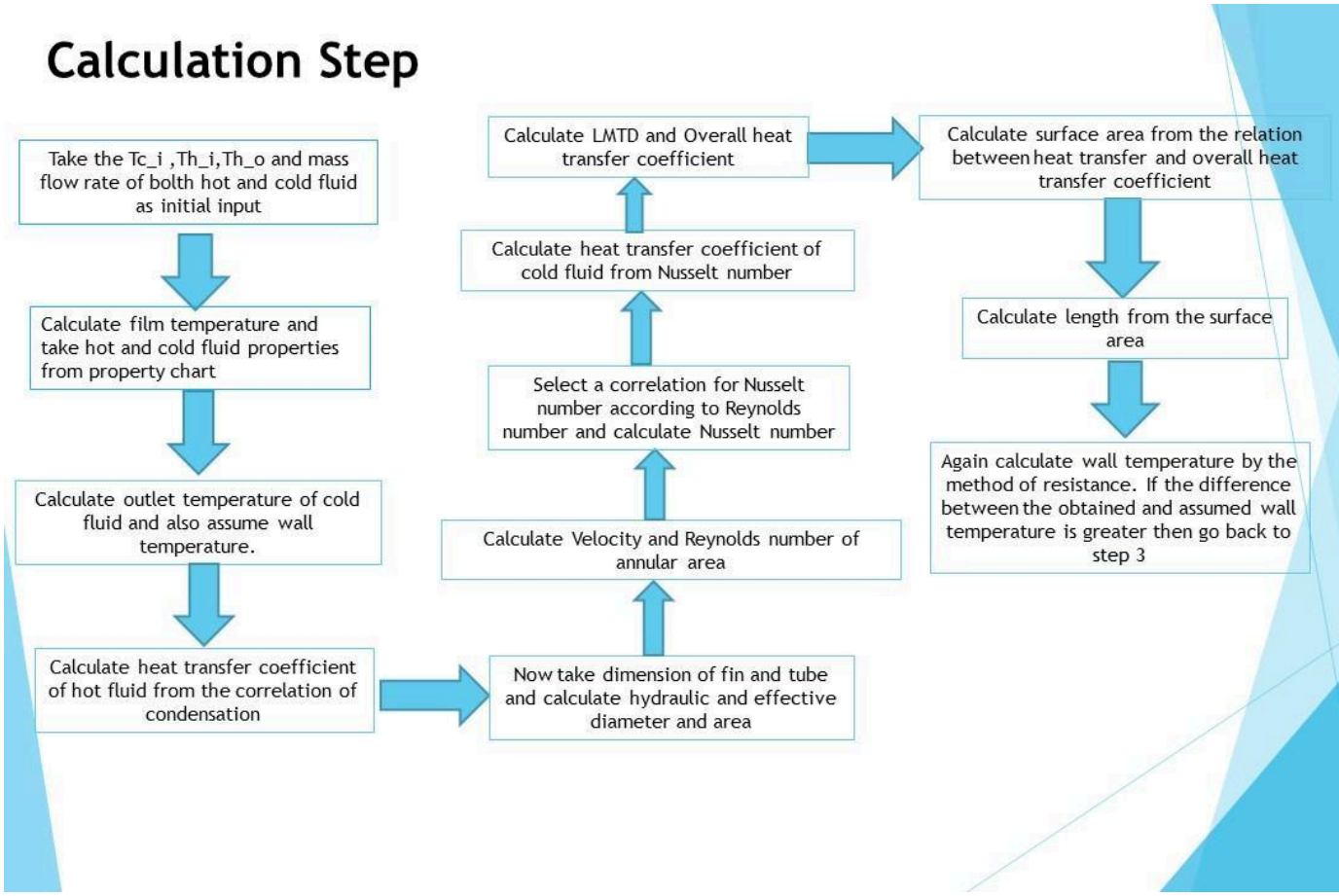
Cross Section



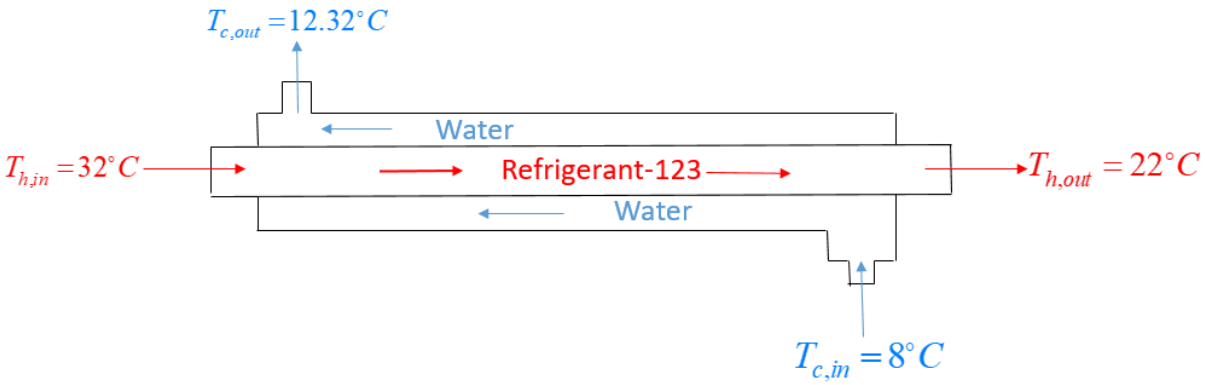




Flow Chart


Calculation Step
Take the Tc_i ,Th_i,Th_o and mass flow rate of bolth hot and cold fluid as initial input
Calculate film temperature and take hot and cold fluid properties from property chart
Calculate outlet temperature of cold fluid and also assume wall temperature.
Calculate heat transfer coefficient of hot fluid from the correlation of condensation
Now take dimension of fin and tube and calculate hydraulic and effective diameter and area
Calculate Velocity and Reynolds number of annular area
Select a correlation for Nusselt number according to Reynolds number and calculate Nusselt number
Calculate heat transfer coefficient of cold fluid from Nusselt number
Calculate LMTD and Overall heat transfer coefficient
Calculate surface area from the relation between heat transfer and overall heat transfer coefficient
Calculate length from the surface area
Again calculate wall temperature by the method of resistance. If the difference between the obtained and assumed wall temperature is greater then go back to step 3

Schematic


$T_{c,out} = 12.32^{\circ}C$
Water
$T_{h,in} = 32^{\circ}C$
Refrigerant-123
$T_{h,out} = 22^{\circ}C$
Water
$T_{c,in} = 8^{\circ}C$

Hand Calculation:

Physical Parameter	Annulus Fluid (Water)	Tube Side Fluid (Refrigerant-123)
Flow rate, kg/s	0.1	0.01
Inlet temperature, °C	8	32
Outlet temperature, °C	-	22
Inner Diameter,(mm)	77.9	32.791
Outer Diameter(mm)	-	34.925

Properties	Annulus Fluid (water)	Tube Side Fluid (Refrigerant-123)
Density at liquid phase, ρ_l (kg/m^3)	990	1459.5
Density at superheated vapor phase, ρ_v (kg/m^3)	–	6.23
Specific Heat of liquid, C_{pl} (kJ/kg. °C)	4194	1.02×10^3
Specific Heat of vapor, C_{pv} (kJ/kg. °C)	–	0.7×10^3
Viscosity of liquid, μ_l (kg/m-s)	1.3×10^{-3}	409.95×10^{-6}
Viscosity of vapor, μ_v (kg/m-s)	-	10.8×10^{-6}
Thermal conductivity of liquid, k_l (W/m-K)	0.58	75.94×10^{-3}
Thermal conductivity of vapor, k_v (W/m-K)	–	9.28×10^{-3}
Prandlt number of liquid, *Pr*	9.45	-

For Tube:

Iteration -1:

Assume,

Surface Temperature, Ts=18°C

The cooling of the liquid below the saturation temperature can be accounted for by replacing hfg by the modified latent heat of vaporization hfg * defined as

$$h_{fg}^{*} = h_{fg} + 0.68C_{pl}(T_{sat} - T_s) + C_{pv}(T_v - T_{sat})$$

$=170.66\times10^3+0.68\times1.02\times10^3(26.7-18)+0.7\times10^3(32-26.7)$

$=180.4\times10^3$ J/Kg

Heat transfer Co-efficient for entire tube,

$$h_i = 0.729\left[\frac{g(\rho_l-\rho_v)h_{fg}^{*}k_l^3}{\rho_l(T_{sat}-T_s)D_{t,i}}\right]^{1/4}$$

$$=0.729\left[\frac{9.8\times1459.5\times(1459.5-6.23)\times180.9\times10^3\times(75.94\times10^{-3})^3}{409.95\times10^{-6}\times(26.7-18)\times32.79\times10^{-3}}\right]^{\frac{1}{4}}$$

$=1412$ W/m^2.K

So, Heat Transfer Rate, $\dot{Q} = \dot{m}_1 h_{fg}^{*}$ $=0.01\times180.4\times10^3=1804$ J

For Annulus Side:

Fin Parameter:

Thickness of fin, δ_f =2mm

Height of fin, H_f =10m

Number of fin, N_f =4

Cross sectional area, $A_c = \frac{\Pi}{4}(D_{a,i}^2 - D_{t,o}^2) - (H_f \; N_f \; t_f)$

$$= \frac{\Pi}{4}(77.9^2 - 34.925^2) - (2 \times 10 \times 4)$$

$$= 3728.12 mm^2$$

Hydraulic Perimeter, $P_{h,a} = \Pi(D_{a,i} + D_{t,o}) + (H_f N_f \times 2)$

$$= \Pi(77.9 + 34.925) + (10 \times 4 \times 2)$$

$$= 434.45 mm$$

Effective Perimeter, $P_{e,a} = \Pi(D_{t,o}) + 2 \times H_f N_f$

$$= \Pi(34.925) + 2 \times 10 \times 4$$

$$= 189.72 mm$$

Hydraulic Diameter, $D_{h,a} = \frac{4A_a}{P_{h,a}} = 0.03432m = 34.32mm$

Effective Diameter, $D_{e,a} = \frac{4A_a}{P_{e,a}} = 0.0786m = 78.6mm$

For outlet Temperature of water,

$$\dot{Q} = \dot{m}_2 C_{pw}(T_{o,w} - 8)$$

$T_{o,w}$ =12.3°C

Velocity, $V_a = \frac{\dot{m}_2}{\rho_w A_a}$

$$= \frac{0.1}{990 \times 3728.12 \times 10^{-6}}$$

$$= 0.027 ms^{-1}$$

Reynold Number, $$R_e = \frac{\rho_w V_a D_{h,a}}{\mu_w}$$

$$= \frac{990 \times 0.027 \times 34.32 \times 10^{-3}}{1.3 \times 10^{-3}}$$

$$= 705.672 \quad \text{(Laminar)}$$

Nusselt Number, $$Nu = 1.86 \times \left[\frac{\mathrm{Re\,Pr}}{D_{t,o}} \right]^{\frac{1}{3}} \times L^{-\frac{1}{3}}$$

$$= 1.86 \times \left[\frac{705.672 \times 9.45}{34.925 \times 10^{-3}} \right]^{\frac{1}{3}} \times L^{-\frac{1}{3}}$$

$$= 107.1 L^{-\frac{1}{3}}$$

Heat Transfer Co-efficient, $$h_o = \frac{N_u K_w}{D_{e,a}}$$

$$= \frac{107.1 L^{-\frac{1}{3}} (0.58)}{78.6 \times 10^{-3}}$$

$$= 790.3 L^{-\frac{1}{3}}$$

Overall Heat Transfer Co-efficient, $$U = \left(\frac{1}{h_i} + \frac{1}{h_o} \right)^{-1}$$

$$= \left(\frac{1}{1412} + \frac{L^{\frac{1}{3}}}{790.3} \right)^{-1}$$

Log Mean Temperature,

$$\Delta T_{LM} = \frac{(32-12.3)-(22-8)}{\ln\left[\frac{32-12.3}{22-8}\right]}$$

$$= 16.688 \text{ °C}$$

Surface Area, $A_s = \pi D_{t,o} N_t L + H_f \times 2$

$$= \pi \times 34.925 \times 10^{-3} \times L + 4 \times L \times 10 \times 10^{-3} \times 2$$

So,

$$UA_s \Delta T_{LM} = \dot{Q}$$

$$\pi \times 34.925 \times 10^{-3} \times L + 4 \times L \times 10 \times 10^{-3} \times 2 = \frac{1804}{16.688} \times \left(\frac{1}{1412} + \frac{L^{\frac{1}{3}}}{790.3} \right)$$

Or,

So, L=1.24 m

Iteration -2:

From Iteration-1 ,

L=1.24m

$\dot{Q}$ =1804J

By the method of conduction,

$$\dot{Q} = \frac{26.7 - T_s}{\frac{1}{h_i A_i}}$$

So, $T_s = 16.69$ °C

$h_{fg}^* = h_{fg} + 0.68C_{pl}(T_{sat} - T_s) + C_{pv}(T_v - T_{sat})$ = 181.3×10^{3}J/Kg

$h_i = 0.729\left[\frac{g(\rho_l - \rho_v)h_{fg}^* k_l^3}{\rho_l(T_{sat} - T_s)D_{t,i}}\right]^{1/4}$ =1364.25 W/m^{2}K

$\dot{Q}$ = $\dot{m}_1 h_{fg}^*$ =1813J

$\dot{Q} = \dot{m}_2 C_{pw}(T_{o,w} - 8)$

$T_{o,w}$ =12.32°C

$\Delta T_{LM} = \frac{(32 - 12.32) - (22 - 8)}{\ln\left[\frac{32 - 12.32}{22 - 8}\right]}$ =16.68°C

Now,

$$\Pi \times 34.925 \times 10^{-3} \times L + 4 \times L \times 10 \times 10^{-3} \times 2 = \frac{1813}{16.68} \times \left(\frac{1}{1364.25} + \frac{\bar{L^3}}{790.3}\right)$$

L=1.26m

Iteration -3:

From Iteration-2 ,

L=1.26m

$\dot{Q}$ =1364.25J

By the method of conduction,

$\dot{Q} = \frac{26.7 - T_s}{\frac{1}{h_i A_i}}$

So, $T_s = 16.46$ °C

$h_{fg}^* = h_{fg} + 0.68C_{pl}(T_{sat} - T_s) + C_{pv}(T_v - T_{sat})$ = 181.4×10^{3}J/Kg

$h_i = 0.729\left[\frac{g(\rho_l - \rho_v)h_{fg}^* k_l^3}{\rho_l(T_{sat} - T_s)D_{t,i}}\right]^{1/4}$ =1356.71 W/m^{2}K

$\dot{Q}$ = $\dot{m}_1 h_{fg}^*$ =1814J

$\dot{Q} = \dot{m}_2 C_{pw}(T_{o,w} - 8)$

$T_{o,w}$ =12.32°C

$\Delta T_{LM} = \frac{(32 - 12.32) - (22 - 8)}{\ln\left[\frac{32 - 12.32}{22 - 8}\right]}$ =16.68°C

Now,

$$\Pi \times 34.925 \times 10^{-3} \times L + 4 \times L \times 10 \times 10^{-3} \times 2 = \frac{1814}{16.68} \times \left(\frac{1}{1356.71} + \frac{\bar{L^3}}{790.3}\right)$$

L=1.27m

Fin Efficiency:

$$\eta_f = \frac{\dot{Q}_{fin}}{\dot{Q}_{actual}}$$

Surface Area for fin, $A_s = N_f L_c H_f \times 2$

$$= 4 \times \left(L + \frac{t}{2}\right) H_f \times 2$$

$$= 4 \times \left(1.27 + \frac{2 \times 10^{-3}}{2} \right) \times 10 \times 10^{-3} \times 2$$

$$= 0.10088 m^2$$

$$h_c = h_o = 790.3 L^{-\frac{1}{3}} = 731.7 W / m^2 K$$

$$\theta_{as} = T_s - T = 16.69 - 10.15 = 6.54°C$$

$$\dot{Q}_{actual} = A_s h_{c\ s}$$

$$= 0.10088 \times 731.7 \times 6.54$$

$$= 482.74 J / s$$

Perimeter of fin, $P = \frac{A_s}{L} = \frac{0.10088}{1.27} = 0.08 m$

Thermal Conductivity, K=384 W/mK

$M = \sqrt{h_o P K A_s} \times$ $_s$ =311.42J

$$m = \sqrt{\frac{h_o P}{K A_s}} = \sqrt{\frac{731.7 \times 0.08}{384 \times 0.10088}} = 1.229$$

$$mL_c = 1.229 \times 1.271 = 1.55$$

$$\frac{h_c}{mK} = 1.55$$

$$\dot{Q}_{fin} = M \times \frac{\tanh(mL_c) + \frac{h_c}{mK}}{1 + \tanh(mL_c) \times \frac{h_c}{mK}}$$

$$= 311.42 \times \frac{\tanh(1.55) + 1.55}{1 + 1.55 \tanh(1.55)}$$

$= 311.42 \times 1.01962$

$= 317.53 J / s$

So, efficiency of the fin, $$\eta_f = \frac{\dot{Q}_{fin}}{\dot{Q}_{actual}} = \frac{317.53}{482.74} \times 100\% = 65.77\% \approx 66\%$$

Pressure Drop:

$$\frac{1}{K} = \frac{D_{i,a}}{D_{o,p}}$$

$$\therefore K = \frac{D_{o,p}}{D_{i,a}} = \frac{34.925}{779} = 0.4483$$

$$\frac{1}{\varepsilon_{corr}} = \frac{1+k^2}{1-k^2}$$

$$\therefore \varepsilon_{corr} = 0.6653$$

$$f = {}_{corr} \times \frac{64}{\mathrm{Re}} = 0.06$$

Pressure Drop in Annulus, $$\Delta P_a = f\left(\frac{2L}{D_{h,a}}\right)\left(\frac{1}{2}\rho V_a^2\right)N_{hp}$$

$$= 0.06\left(\frac{2 \times 1.27}{34.32 \times 10^{-3}}\right)\left(\frac{1}{2} \times 990 \times 0.027^2\right) \times 1$$

$$= 1.589 Pa$$

Calculation without Fin:

Annular Area: $$A_a = \frac{\Pi}{4}\left(D_{a,i}^2 - D_{a,o}^2\right)$$

$$= \frac{\Pi}{4} \times \left(77.9^2 - 34.925^2\right)$$
$$= 3808.12mm^2$$

Hydraulic Perimeter: $P_{h,a} = \Pi \left(D_{a,i} + D_{t,o}\right)$

$$= \Pi \left(77.9 + 34.925\right)$$
$$= 354.45mm$$

Effective Perimeter: $P_{e,a} = \Pi \times D_{t,o}$

$$= \Pi \times 34.925$$
$$= 109.72mm$$

Hydraulic Diameter: $D_{h,a} = \frac{4A_a}{P_{h,a}} = 42.97mm$

Effective Diameter: $D_{e,a} = \frac{4A_a}{P_{e,a}} = 138.83mm$

Velocity of water: $V_a = \frac{\dot{m}_2}{\rho_w \times A_a}$

$$= \frac{0.1}{990 \times 3808.12 \times 10^{-6}}$$
$$= 0.02652ms^{-1}$$

Reynold number: $R_e = \frac{\rho_w V_a D_{h,a}}{\mu_w}$

$$= \frac{990 \times 0.02652 \times 42.97 \times 10^{-3}}{1.3 \times 10^{-3}}$$
$$= 867.82$$

Nusselt number: $$Nu = 1.86 \times \left[\frac{\mathrm{Re}\,\mathrm{Pr}}{D_{t,o}} \right]^{\frac{1}{3}} \times L^{-\frac{1}{3}}$$

$$= 1.86 \left(\frac{867.82 \times 9.45}{34.925 \times 10^{-3}} \right)^{1/3} \times L^{-1/3}$$

$$= 114.75 L^{-1/3}$$

Determining h_o:

$$h_o = \frac{NuK_w}{D_{e,a}} = \frac{114.75 L^{-1/3} \times 0.58}{138.83 \times 10^{-3}} = 480 L^{-1/3}$$

Overall heat transfer Coefficient, $$U = \left(\frac{1}{h_o} + \frac{1}{h_i} \right)^{-1}$$

$\dot{Q} = UA\,\Delta T_{LM}$

$$\Rightarrow \quad \Pi \times 31.925 \times 10^{-3} L + 4L \times 10 \times 10^{-3} \times 2 = \frac{1804}{16.68} \left(\frac{1}{1412} + \frac{L^{1/3}}{480} \right)$$

$$\therefore L = 4.46m$$

Here with fin the required length decreases to 1.27m. Thus we can conclude that the addition of longitudinal fins in a double pipe enhances heat transfer efficiency by increasing the surface area available for heat exchange. This improved efficiency reduces the required length of the pipe, leading to material savings and compact system design. Additionally, it optimizes energy utilization and improves overall thermal performance.

FINAL RESULT (From hand calculation:)

Length of the double pipe,L=1.27m

Overall heat transfer Coefficient,U=474.5 W/m^2.K

Fin efficiency, η_f =66%

Pressure Drop in Annulus=1.589Pa

HTRI RATING

HTRI

HEAT EXCHANGER RATING DATA SHEET

Released to the following company:
BUET
Dilshad Jahan Ritu

SI Units

Service of Unit — Item No.

Type: Hairpin — Orientation: Horizontal — Connected In: 1 Parallel 1 Series

Surf/Unit (Gross/Eff): 0.646 / 0.524 m2 — Shell/Unit: 1 — Surf/Shell (Gross/Eff): 0.646 / 0.524 m2

PERFORMANCE OF ONE UNIT

Fluid Allocation		Shell Side		Tube Side	
Fluid Name		water		Refrigerant-123	
Fluid Quantity, Total	kg/s	0.1000		0.01	
Vapor (In/Out)	wt%	0.00	0.00	100.00	0
Liquid	wt%	100.00	100.00	0.00	100
Temperature (In/Out)	C	8.00	12.83	32.00	22.00
Density	kg/m3	990.00	990.00	6.23	1459.5
Viscosity	mN-s/m2	1.3000	1.3000	0.0108	0.4101
Specific Heat	kJ/kg-C	4.1940	4.1940	0.7000	0.99
Thermal Conductivity	W/m-C	0.5800	0.5800	0.0093	0.07431
Critical Pressure	kPa				
Inlet Pressure	kPa	101.00		101.00	
Velocity	m/s		2.71e-2		0.81
Pressure Drop, Allow/Calc	kPa		0.044		0.820
Average Film Coefficient	W/m2-K	100.35		980.59	
Fouling Resistance (min)	m2-K/W	0.000350		0.000350	

Heat Exchanged: 0.0070 MegaWatts — MTD (Corrected): 10.5 C — Overdesign: 4.23 %

Transfer Rate, Service: 1269.3 W/m2-K — Calculated: 77.62 W/m2-K — Clean: 84.11 W/m2-K

CONSTRUCTION OF ONE SHELL

Sketch (Bundle/Nozzle Orientation)



			Shell Side	Tube Side
Design Pressure		kPaG	517.11	517.11
Design Temperature		C	5.6	35
No Passes per Shell			1	1
Flow Direction			Upward	
Connections Size & Rating	In	mm	1 @ 26.645	1 @ 40.894
	Out	mm	1 @ 26.645	1 @ 40.894
	Liq. Out	mm	@	1 @

Tube No. 1.0000 OD 34.925 mm — Thk(Avg) 1.067 mm — Length 1.270 m — Pitch 43.656 mm — Tube pattern 30

Tube Type: Longitudinal Fin — Material: Copper — Pairs seal strips: 1

Shell ID: 77.900 mm — Kettle ID: mm — Passlane Seal Rod No.: 0

Cross Baffle Type: None — %Cut (Diam) — Impingement Plate: None

Spacing(c/c): 2760.4 mm — Inlet: mm — No. of Crosspasses: 1

Rho-V2-Inlet Nozzle: 32.49 kg/m-s2 — Shell Entrance: 3.12 kg/m-s2 — Shell Exit: 3.12 kg/m-s2

Bundle Entrance: 0.00 kg/m-s2 — Bundle Exit: 0.00 kg/m-s2

Weight/Shell: 154.60 kg — Filled with Water: 168.99 kg — Bundle: 13.89 kg

Notes:

Thermal Resistance, %		Velocities, m/s		Flow Fractions	
Shell	77.35	Shellside	2.71e-2	A	
Tube	14.58	Tubeside	0.81	B	1.000
Fouling	7.72	Crossflow	0.00	C	0.000
Metal	0.35	Window	2.71e-2	E	
				F	

Output Summary

Released to the following company:
BUET
Dilshad Jahan RITU

Xhpe 9 (32 bit) 12/9/2024 10:18 SN: 46639- **SI Units**

Shell 1
Rating - Horizontal Countercurrent Flow Hairpin

See Data Check Messages Report for Warning Messages.
See Runtime Message Report for Warning Messages.

Process Conditions		Cold Shellside		Hot Tubeside	
Fluid name			water		Refrigerant-123
Total flow rate	(kg/s)		0.1000		0.01
Inlet/Outlet Y	(Wt. frac vap.)	0.0000	0.0000	1.0000	1.0000 *
Inlet/Outlet T	(Deg C)	8.00	12.83	32.00	22.00
Inlet P/Avg	(kPa)	101.00	100.98	101.00	100.59
dP/Allow.	(kPa)	0.044	0.000	0.820	0.000
Fouling	(m2-K/W)		0.000350		0.000350

Exchanger Performance

Shell h	(W/m2-K)	820.8	Actual U	(W/m2-K)	492.1
Tube h	(W/m2-K)	1230	Required U	(W/m2-K)	480
Hot regime	(--)	Sens. Gas	Total duty	(MegaWatts)	0.0021
Cold regime	(--)	Sens. Liquid	Eff. area	(m2)	0.256
EMTD	(Deg C)	10.5	Overdesign	(%)	4.23

Shell Geometry			Baffle Geometry		
Type	(--)	Hairpin	Baffle type		None
Shell ID	(mm)	77.900	Baffle cut	(Pct Dia.)	
Series	(--)	1	Baffle orientation	(--)	
Parallel	(--)	1	Central spacing	(mm)	2760.4
Orientation	(deg)	0.00	Crosspasses	(--)	1

Tube Geometry			Nozzles		
Tube type	(--)	Longitudinal Fin	Shell inlet	(mm)	26.645
Tube OD	(mm)	34.925	Shell outlet	(mm)	26.645
Nominal length	(m)	1.270	Inlet height	(mm)	21.487
Pitch ratio	(--)	1.2500	Outlet height	(mm)	21.487
Layout	(deg)	30	Tube inlet	(mm)	40.894
Tubecount	(--)	1	Tube outlet	(mm)	40.894

Thermal Resistance, %	
Shell	77.35
Tube	14.58
Fouling	7.72
Metal	0.35

Velocities, m/s			
		Min	Max
Tubeside		0.82	0.81
Crossflow		--	--
Longitudinal		2.71e-2	2.71e-2

Flow Fractions	
A	
B	1.000
C	0.000
E	
F	

Xhpe 9 (32 bit) 12/9/2024 10:18 SN: 46639- **SI Units**

Shell 1
Rating - Horizontal Countercurrent Flow Hairpin

Point number	(--)	1	2	3	4	5	6
Tube Pass	(--)	1	1	1	1	1	1
Length from tube inlet	(mm)	0.000	138.022	414.06	690.1	966.16	1.2622
Mass fraction vapor	(--)	1.0000	0.791	0.553	0.432	0.234	0
Bulk temperature	(C)	32.00	31.22	30.4	28.93	27.54	25.89
Skin temperature	(C)	31.22	30.69	30.19	29.69	29.18	28.66
Wall metal temperature	(C)		30.31	29.80	29.28	28.76	28.23
MTD	(C)		7.5	7.8	8.1	8.4	8.7
Superheat for Onset of Nucleate Boiling	(C)						
Pressure	(kPa)	101.00	100.88	100.85	100.81	100.78	100.75
Pressure drop	(kPa)	0.104	0.032	0.032	0.032	0.032	0.032
Enhanced pressure drop mult	(--)						
Friction loss	(kPa)		0.032	0.032	0.032	0.032	0.032
Static head loss	(kPa)		0.000	0.000	0.000	0.000	0.000
Momentum loss	(kPa)		0.000	0.000	0.000	0.000	0.000
Maximum velocity	(m/s)		0.81	0.81	0.81	0.81	0.81
Average velocity	(m/s)		0.81	0.81	0.81	0.81	0.81
Shear stress at tube wall	(Pa)		0.00	0.00	0.00	0.00	0.00
Area	(m2)		0.026	0.026	0.026	0.026	0.026
Cumulative area	(m2)		0.026	0.052	0.079	0.105	0.131
Duty	(MegaWatts)		3.16e-4	3.19e-4	3.23e-4	3.26e-4	3.30e-4
Cumulative duty	(MegaWatts)	0.0000	3.16e-4	6.35e-4	9.58e-4	0.0013	0.0016
Heat flux	(W/m2)		12054	12189	12324	12460	12597
Calculated heat flux	(W/m2)		578.99	602.53	626.32	650.38	674.71
Critical heat flux	(W/m2)						
Overall U	(W/m2-K)		492.1	492.1	492.1	492.1	492.1
Tubeside h	(W/m2-K)		1230.05	1230.05	1230.05	1230.05	1230.05
Enhanced heat transfer mult	(--)						
Sensible liquid h	(W/m2-K)		0.00	0.00	0.00	0.00	0.00
Sensible vapor h	(W/m2-K)		1230.05	1230.05	1230.05	1230.05	1230.05
Condensate film h	(W/m2-K)		0.00	0.00	0.00	0.00	0.00
Vapor phase h	(W/m2-K)		1230.05	1230.05	1230.05	1230.05	1230.05
Nucleate boiling h	(W/m2-K)		0.00	0.00	0.00	0.00	0.00
Conv. boiling h	(W/m2-K)		0.00	0.00	0.00	0.00	0.00
Film boiling h	(W/m2-K)		0.00	0.00	0.00	0.00	0.00
Boiling thin film h	(W/m2-K)		0.00	0.00	0.00	0.00	0.00
Natural convective h	(W/m2-K)		0.00	0.00	0.00	0.00	0.00
Local Reynolds	(--)	3593920	3595406	3595406	3595406	3595406	3595406
Vapor Reynolds	(--)	3593920	3595406	3595406	3595406	3595406	3595406
Liquid Reynolds	(--)						
Vapor Prandtl	(--)	0.8155	0.8155	0.8155	0.8155	0.8155	0.8155
Liquid Prandtl	(--)	5.5118	5.5118	5.5118	5.5118	5.5118	5.5118
Grashof	(--)		2.3e+10	1.9e+10	1.9e+10	2.0e+10	2.1e+10
Richardson	(--)		0.0018	0.0014	0.0015	0.0016	0.0016
Flow regime param.	(--)		0.0000	0.0000	0.0000	0.0000	0.0000
Condensate regime	(--)		Sens Gas	Sens Gas	Sens Gas	Sens Gas	Sens Gas
Boiling flow regime	(--)						
Boiling mechanism	(--)						

Xhpe 9 (32 bit) 12/9/2024 10:18 SN: 46639- **SI Units**

Shell 1
Rating - Horizontal Countercurrent Flow Hairpin

Shellside Flow Region	**1**	0.1000 kg/s					
Point number	(--)	1	2	3	4	5	6
Shell pass	(--)	1	1	1	1	1	1
Length from tube inlet	(mm)	1270	1230.6	1160.9	960.56	770.89	690.26
Mass fraction vapor	(--)	0	0.0000	0.0000	0.0000	0.0000	0.0000
Bulk temperature	(C)	8.00	8.46	9.37	10.28	10.99	11.18
Skin temperature	(C)		14.23	14.89	15.25	15.96	16.41
Wall metal temperature	(C)		14.23	14.91	15.25	15.97	16.42
MTD	(C)		13.8	13.4	13.1	12.7	12.4
Superheat for Onset of Nucleate Boiling	(C)						
Delta MTD correction	(--)		1.0000	1.0000	1.0000	1.0000	1.0000
Pressure	(kPa)	101.00	100.98	100.98	100.98	100.98	100.98
Pressure drop	(kPa)	0.020	1.32e-4	1.32e-4	1.32e-4	1.32e-4	1.32e-4
Friction loss	(kPa)		1.32e-4	1.32e-4	1.32e-4	1.32e-4	1.32e-4
Static head loss	(kPa)		0.000	0.000	0.000	0.000	0.000
Momentum loss	(kPa)		0.000	0.000	0.000	0.000	0.000
Crossflow velocity	(m/s)						
Longitudinal velocity	(m/s)		2.71e-2	2.71e-2	2.71e-2	2.71e-2	2.71e-2
Shear stress at tube wall							
Crossflow	(Pa)		0.00	0.00	0.00	0.00	0.00
Longitudinal flow	(Pa)		1.49e-2	1.49e-2	1.49e-2	1.49e-2	1.49e-2
Area	(m2)		0.026	0.026	0.026	0.026	0.026
Cumulative area	(m2)		0.026	0.052	0.079	0.105	0.131
Duty	(MegaWatts)		3.86e-4	3.82e-4	3.78e-4	3.74e-4	3.71e-4
Cumulative duty	(MegaWatts)		3.86e-4	7.68e-4	0.0011	0.0015	0.0019
Heat flux	(W/m2)		14736	14588	14441	14295	14150
Calculated heat flux	(W/m2)		1072.3	1043.9	1015.7	987.78	960.17
Critical heat flux	(W/m2)		0.00	0.00	0.00	0.00	0.00
Overall U	(W/m2-K)		492.1	492.1	492.1	492.1	492.1
Shellside h	(W/m2-K)		820.8	820.8	820.8	820.8	820.8
Sensible liquid h	(W/m2-K)		820.8	820.8	820.8	820.8	820.8
Sensible vapor h	(W/m2-K)		0.00	0.00	0.00	0.00	0.00
Condensate film h	(W/m2-K)		0.00	0.00	0.00	0.00	0.00
Vapor phase h	(W/m2-K)		0.00	0.00	0.00	0.00	0.00
Nucleate boiling h	(W/m2-K)		0.00	0.00	0.00	0.00	0.00
Conv. boiling h	(W/m2-K)		0.00	0.00	0.00	0.00	0.00
Film boiling h	(W/m2-K)		0.00	0.00	0.00	0.00	0.00
Boiling thin film h	(W/m2-K)		0.00	0.00	0.00	0.00	0.00
Natural convective h	(W/m2-K)		0.00	0.00	0.00	0.00	0.00
Local Reynolds	(--)	2693.1	1621.9	1621.9	1621.9	1621.9	1621.9
Vapor Reynolds	(--)						
Liquid Reynolds	(--)	2693.1	1621.9	1621.9	1621.9	1621.9	1621.9
Vapor Prandtl	(--)						
Liquid Prandtl	(--)	9.4097	9.4058	9.4058	9.4058	9.4058	9.4058
Grashof number	(--)		0.00000	0.00000	0.00000	0.00000	0.00000
Richardson number	(--)		0.0000	0.0000	0.0000	0.0000	0.0000
Flow regime param.	(--)		0.0000	0.0000	0.0000	0.0000	0.0000
Condensate regime	(--)						
Boiling flow regime	(--)		Sens Liq	Sens Liq	Sens Liq	Sens Liq	Sens Liq
Boiling mechanism	(--)		Sens Liq	Sens Liq	Sens Liq	Sens Liq	Sens Liq

Observation: It is ovserved that the result of our calculation is very close to the value we've got from HTRI.

Pressure Testing

After making the prototype, it was taken to BUET Hydraulics Lab for pressure testing. To test the pressure, the outlet of the copper tube was closed by a dead cap after the tubes were filled with water. And then in the inlet water pressure was being increased using a hand pump. And the amount of pressure given was seen from the pressure gauge attached with the hand pump. And the lab officials confirmed before the test, that the pressure gauge was calibrated properly. The pressure could be given is slightly greater than 300 psi . So, it's confirmed that our prototype can withstand at least a pressure of 350 psi. With more safety measurements, and other equipment maybe checked the maximum pressure the prototype can withstand before failing. From the pressure testing, we can come to the conclusion that the selection of materials and joining process was good enough and there is no leakage found.

Section-B2 : Group-01

2010101 → Mostafizur Rahman
2010102 – Dilshad Jahan Ritu
2010106 → Md Abir Khan
2010110 – Rifat Bin Mamun

Pressure test

350 Psi Sustained

Titu
Lab Instructor
ME, BUET

Annulus (5 ft)	1600
Socket	250
Welding	3500
Reducer	70
Painting	560
Insulation	30
Total	7510

Cost Analysis